LECTURE 14

LAST TIME

LECTURE 13: REVIEW

Black–Scholes formula!

THIS TIME

LECTURE 14: OUTLINE

Greeks

DELTA

Let V be a derivative

Delta of V tells us how much V changes for a small change in S_t

$$\Delta := \frac{\partial V}{\partial S}$$

REMINDER: HEDGING CALLS AND PUTS

Is the delta of a call positive or negative?

What are its maximum and minimum values?

Is the delta of a put positive or negative?

What are its maximum and minimum values?

HEDGE RATIO

The delta is also called the hedge ratio

Suppose B–S holds, so

$$c_t = S_t \Phi(d_1) - e^{-r(T-t)} K \Phi(d_2)$$

How many shares do you hold in the replicating portfolio?

HOMEWORK EXERCISE

Try to derive delta from the BS formula

You will see that

$$\Delta \equiv \frac{\partial V}{\partial S} = \Phi(d_1)$$

GAMMA

Gamma is the essence of optionality

Measures how much the delta will change for small change in S

$$\Gamma := \frac{\partial^2 V}{\partial S^2} = \frac{\partial \Delta}{\partial S}$$

If B–S holds, you can compute the gamma of a call as

$$\Gamma = \frac{\phi(d_1)}{S\sigma\sqrt{T-t}}$$

GAMMA MEASURES ADJUSTMENT RATE

You are hedging a call has a gamma of .12

You hold a call on 100 shares

If S increases by \$1, how many shares do you buy to stay hedged?

GAMMA- AND DELTA-HEDGING

Constantly changing stock prices lead to changing deltas

In practice, transactions costs make continuous hedging expensive

Gamma measures how fast you have to change hedging portfolio

Gamma is typically hedged with other options

VEGA

VEGA

Vega ν (sometimes called kappa too) is the sensitivity of V to σ,

$$\nu := \frac{\partial V}{\partial \sigma}$$

In B–S, you can compute

$$\nu_{\text{call}} = \nu_{\text{put}} = S\phi(d_1)\sqrt{T-t}$$

Formula leads traders to quote prices in "vol" instead of dollars

Since $\nu > 0$, more vol means higher price

WHY IS VEGA POSITIVE?

To see why vega is positive, use an option price decomposition

$$\text{option price} \;=\; \text{value of forward} \;+\; \text{insurance value}$$

E.g. for a call

$$c_t = S_t - e^{-r(T-t)}K + \text{ insurance value}$$

The value of the forward doesn't depend on vol

But the insurance value increases as vol increases

Thus option prices increase in vol or $\nu > 0$

THETA

THETA

Even if stock prices don't change, time passing creates P&L

The "time bias" of a position is effect of decreasing t.t.m. on price

For derivative V

If V_t decreases as t.t.m. decreases, time bias is negative

If V_t increases as t.t.m. decreases, time bias is positive

Calls and puts usually "bleed" (losing optionality)

Define theta to measure the time bias

$$\Theta = \frac{\partial V}{\partial t} = -\frac{\partial V}{\partial (T-t)}$$

THETA FOR B–S CALLS

For B–S calls, you can compute that

$$\Theta = -re^{-r(T-t)}K\Phi(d_2) - \frac{S\sigma\phi(d_1)}{2\sqrt{T-t}} < 0$$

$T - t$ goes with either r or σ

DISCOUNT AND VOL EFFECTS

Reminiscent of time value decomposition

With r: pure discount effect

Smaller t.t.m., less K is discounted, the more to be paid

You pay more, so the option is worth less

With σ: volatility effect

$\sigma\sqrt{T-t}$ is the vol to maturity, so higher $T-t$ is high vol

Thus, higher price

For puts, theta can be positive due to positive discount effect

RHO

RHO

Rho measures sensitivity of V to changes in the short rate

Positive for a call, negative for a put

In the B–S world, you can compute

$$\rho_{\text{call}} = K(T-t)e^{-r(T-t)}\Phi(d_2)$$

and

$$\rho_{\text{put}} = -K(T-t)e^{-r(T-t)}\Phi(-d_2)$$

BOOKS

BOOKS

IBanks don't hedge position-by-position

They hedge their entire book in one go

They find the greeks of the entire book to hedge them

This makes writing options cheaper and saves delta hedging costs

Traders of course don't hedge the entire book perfectly

They may find it expensive to hedge part of it

or wish to leave some unhedged for speculative reasons

RECAP

LECTURE 14: RECAP

The Greeks!

FIN